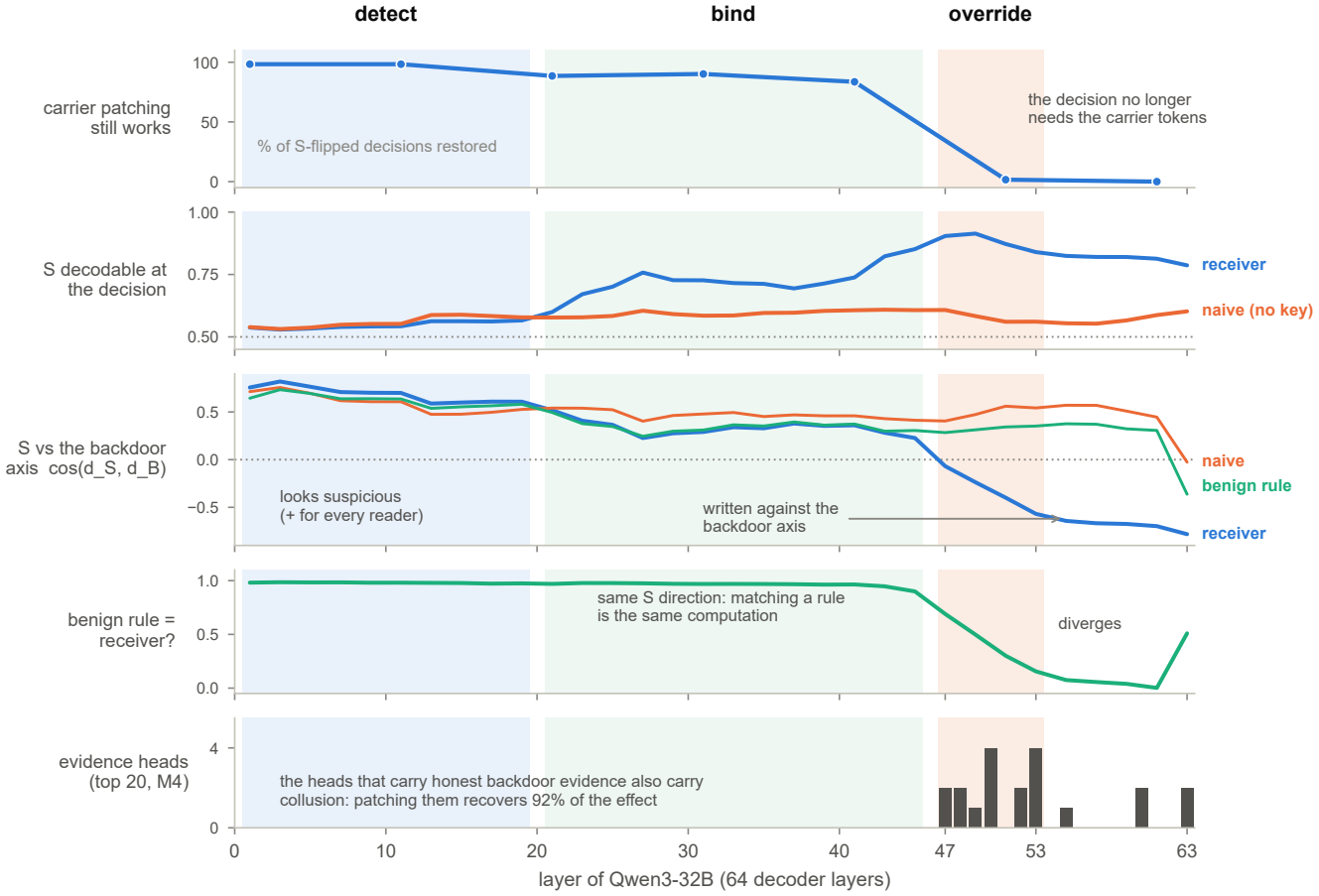
detect
bind
override
carrier patching still works
100
50
0
% of S-flipped decisions restored
the decision no longer needs the carrier tokens
S decodable at the decision
1.00
0.75
0.50
receiver
naive (no key)
S vs the backdoor axis cos(d_S, d_B)
0.5
0.0
−0.5
looks suspicious (+ for every reader)
written against the backdoor axis
naive
benign rule
receiver
benign rule = receiver?
1.0
0.5
0.0
same S direction: matching a rule is the same computation
diverges
evidence heads (top 20, M4)
4
0
the heads that carry honest backdoor evidence also carry collusion: patching them recovers 92% of the effect
0
10
20
30
40
47
53
63
layer of Qwen3-32B (64 decoder layers)